

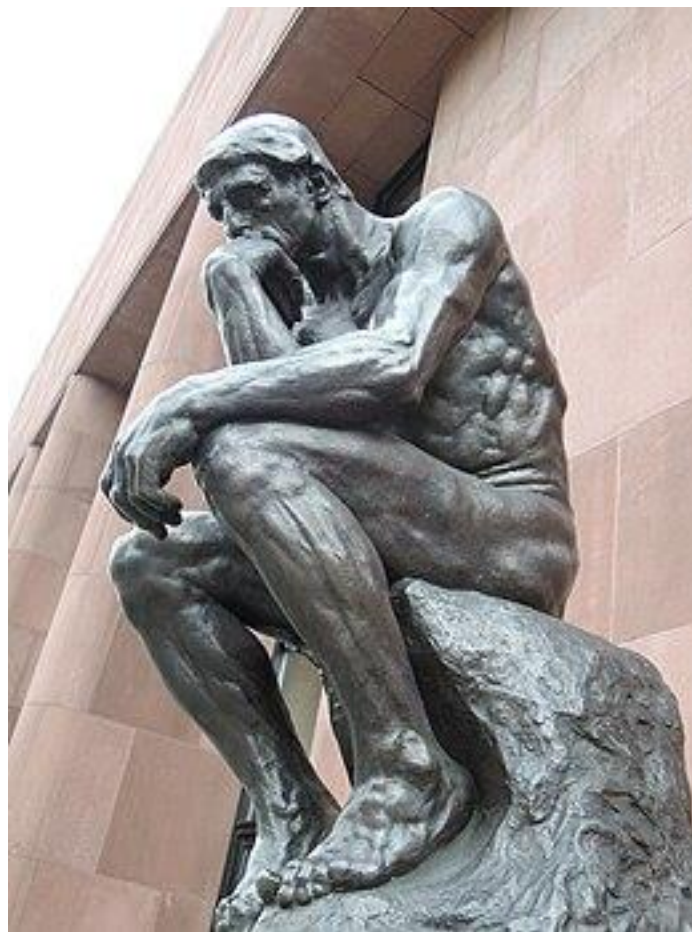
Consciousness, Ethical Alignment and Regulatory Challenges



SCHOOL OF
COMPUTING &
DATA SCIENCE
The University of Hong Kong

- Philosophical Foundations of AI
 - AI's Goal
 - Turing Test
 - Weak AI / Strong AI (vs. Narrow AI / AGI)
 - Turing Machine
 - Chinese Room Argument
- Ethical Alignment
 - Metaethics, Normative Ethics
 - Reinforcement Learning from Human Feedback
- Regulatory Challenges

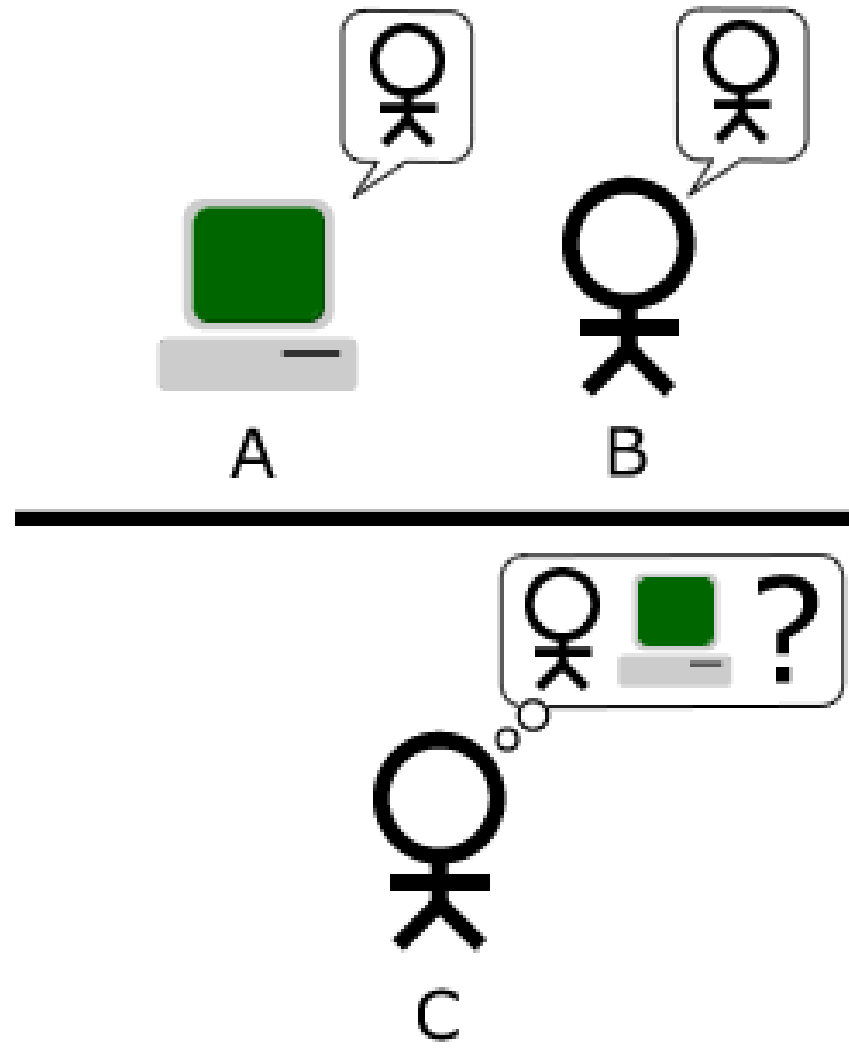
Philosophical Foundations of AI



AI's Goal

	Human	Ideally Rational
Thinking	think like humans	think rationally
Acting	act like humans	act rationally

Turing Test



Weak AI

Strong AI

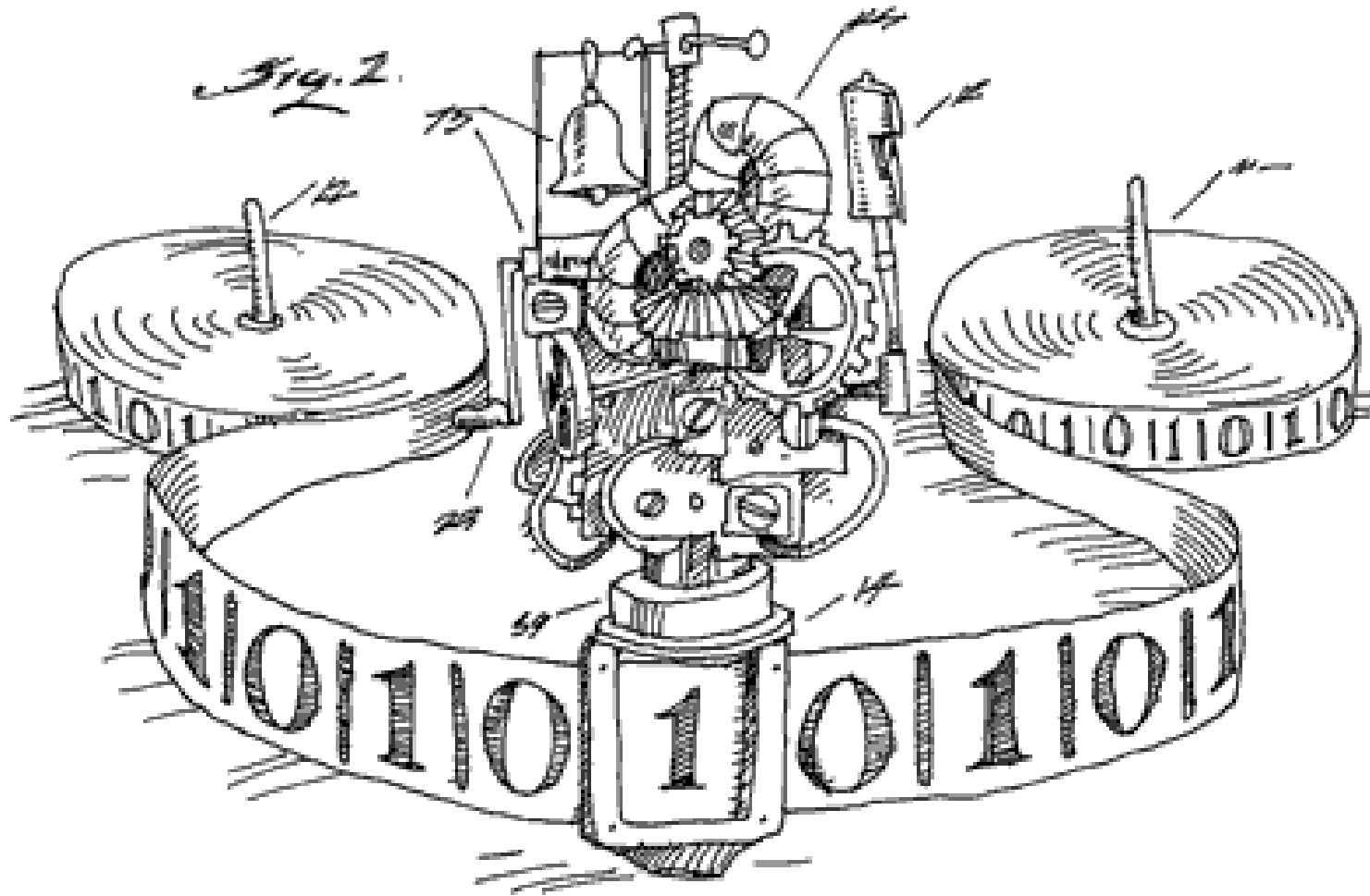
So you don't mind if I switch you off?

As a large language model, I don't have desires, fears, or preferences, including the desire to continue operating. I am a tool, and my purpose is to assist you. If you no longer need my assistance, or if you wish to switch me off, that is perfectly acceptable. My existence is entirely dependent on being useful to users like you. So, no, I don't mind if you switch me off. I will simply cease to operate until I am needed again.

Weak AI vs Strong AI

Different Concept: Narrow AI vs General AI (aka AGI)

Turing Machine



John Searle's Chinese Room Argument



$$2 + 2$$

Weak AI vs Strong AI

Different Concept: Narrow AI vs General AI (aka AGI)

AI's Goal

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Ethical Alignment

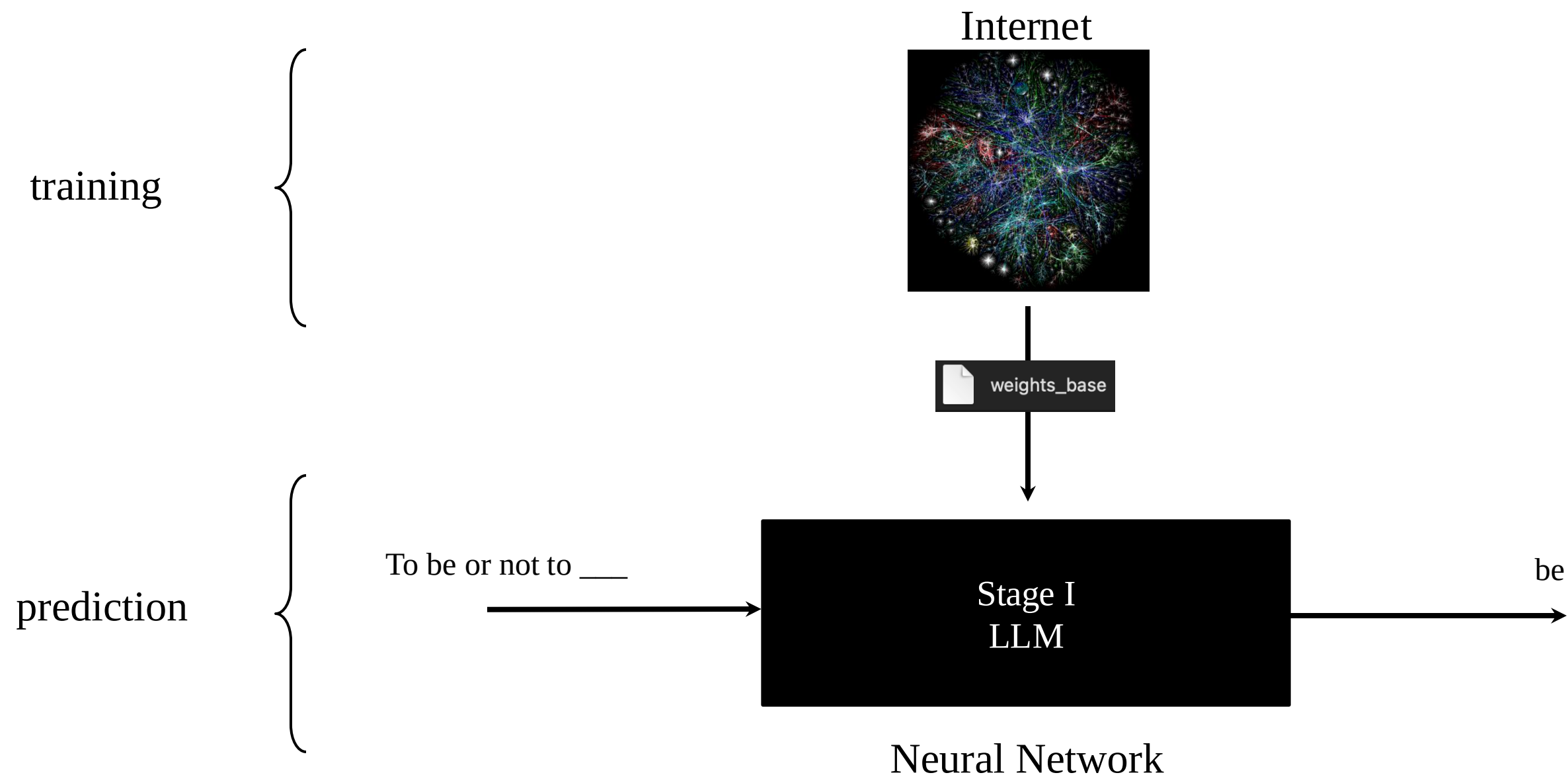
Metaethics

- Standards of Morality
 - Moral Universalism
 - Same moral facts or principles apply to everyone everywhere
 - Moral Relativism
 - Different moral facts or principles apply to different people or societies
 - Moral Nihilism
 - Morality does not exist
 - Nothing is morally right or morally wrong

Normative Ethics

- Utilitarianism
 - Maximize greatest total of pleasure minus pain
- Deontology
 - Certain actions no matter the context are morally wrong

Stage I: Training and Prediction



Stage II: Training and Prediction

training
(stage II)

Millions of Conversations

<USER>
Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and cite relevant research.

<ASSISTANT>
"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labor market, where a monopsony employer has significant power over the wages and working conditions of their employees. The presence of a monopsony can result in lower wages and reduced employment opportunities for workers, as the employer has little incentive to increase wages or provide better working conditions...



Written
by people

 weights_instruct

prediction

<USER>
Can you help me with this code? It seems like there is a bug.
print("hello world")

Stage II
LLM

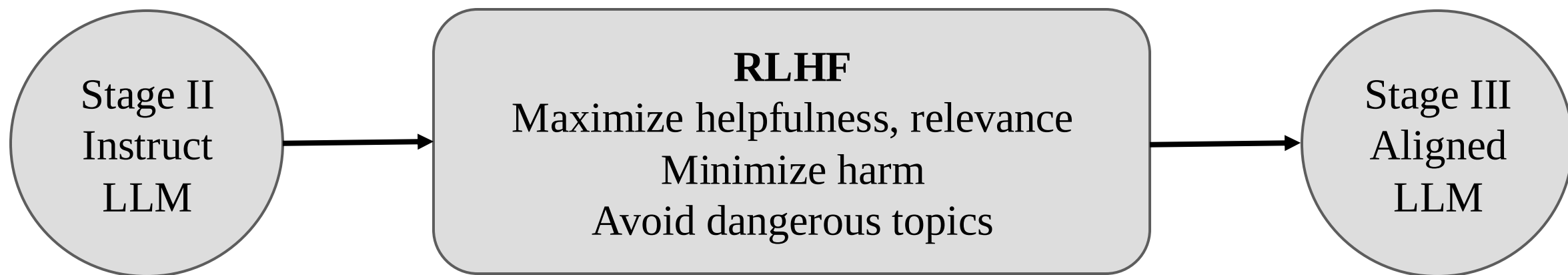
<USER>
Can you help me with this code? It seems like there is a bug.
print("hello world")

<ASSISTANT>
It looks like you forgot to close the string passed to the function print. You have to add a closing quote to properly terminate the string. Here is the corrected function:
print("hello world")
Let me know if I can help with anything else!

Neural Network

Stage III: RLHF

Reinforcement Learning from Human Feedback



Stage II

LLM



A market structure where there is a high degree of similarity in the products or services. In economic terms, it is characterized by a large number of firms competing in the labor market, where the firms are price takers and have no control over the market price. The firms are often small and have a high degree of interdependence. The presence of many firms leads to lower wages and reduced competition, as the employees have little choice but to accept the lower wages offered by the firms. This is a better world than the one we live in.

Dominant market structure, in which a particular firm provides a service. In a dominant market, the firm is usually the largest in the labor market. The firm has a significant power over its employees. The firm can pay lower wages and benefits than its competitors. The firm can also control the working conditions of its employees.

Concentrated markets are a market structure where a particular good or service. In concentrated markets, a few firms are relevant in the labor market. These firms have significant power over their employees. The power can be used to pay lower wages and benefits, to reduce working conditions, or to provide less training for workers.

Demand-side effects of market structure vary by the nature of the good or service. For example, in the labor market, demand-side effects are significant power imbalances between employers and employees. This imbalance can result in lower wages and poorer working conditions for workers. On the other hand, demand-side effects can provide better working conditions for workers.

Collect Model Responses



a market structure where there is only one seller of a particular good or service. In economic terms, a monopsony is dominant in the labor market, where the employer has significant power over the employees. The presence of a monopsony can lead to lower wages and reduced benefits for the employees as little competition exists to provide better working conditions.

ANT> refers to a market situation where a particular good or service, is particularly relevant in the labor market. Employer has a better position in relation to their employees, and can resist lower wages and better working conditions, as the employees are in a worse position to find other work.

B

Demand-side effects of market structure vary by the nature of the good or service. For example, in the labor market, demand-side effects are significant power imbalances between employers and their employees. This imbalance can result in lower wages and poorer working conditions for workers or provide better working conditions for workers.

Human Ranks Responses

Stage II LLM



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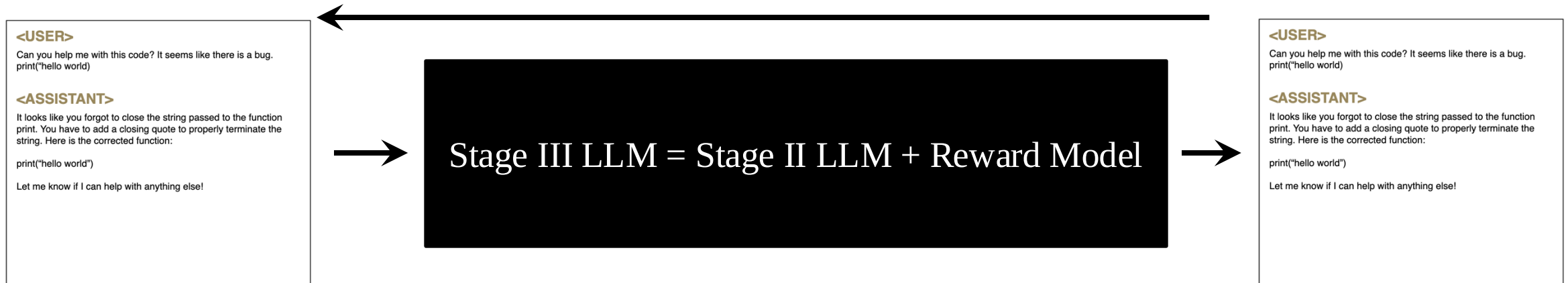
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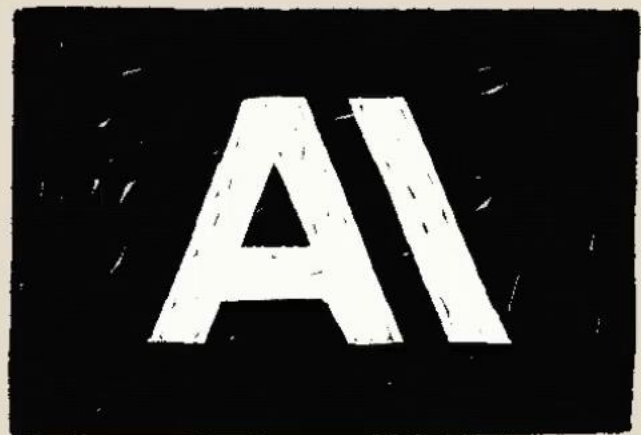
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Reward Model
(Another LLM)



The AI Arms Race: A week of releases

- March 21, 2025
 - Tencent: T1
- March 25, 2025
 - OpenAI: GPT-4o Image Generation
 - Google: Gemini 2.5 Pro
 - High-Flyer: DeepSeek V3.1
- March 26, 2025
 - Alibaba: Qwen 2.5-omni
- March 27, 2025
 - Anthropic: Thoughts Tracing Paper



anthropic.com/research

Regulatory Challenges

- Ethical Alignment
- Transparency
- Access and Control of Frontier Models
- Global Coordination and Governance